

CS1: Shipbuilding Production Scheduling Optimization

You are a junior production planner at a shipyard that builds cargo vessels using modular construction. The shipyard constructs ships from prefabricated sections called "blocks" that are manufactured through a series of production stages. **Your task is to develop an optimal production schedule using CPLEX CPO** through IBM's DOcplex Python API.

The shipyard needs to schedule the production of 24 hull blocks through 4 sequential stages using specialized equipment. You will progressively add complexity to your optimization model, learning fundamental concepts in production scheduling.

Learning Objectives

By completing this exercise, you will:

- Understand basic constraint programming concepts
- Learn to use DOcplex Python API for optimization
- Apply scheduling optimization to real manufacturing problems
- Experience trade-offs between different objectives
- Develop practical problem-solving skills

Problem Context: Simplified Production System

- **24 blocks** across 3 types: Bottom Blocks (BB), Side Blocks (SB), Deck Blocks (DB)
- **4 sequential stages**: Steel Cutting → Welding → Assembly → Finishing
- **8 machines** distributed across the 4 stages
- **Deterministic processing times** (no uncertainty)
- **No quality issues** (simplified manufacturing)

Data Provided

You will receive an Excel file containing:

- Block specifications and quantities
- Processing times for each block type on each machine
- Setup times between different block types
- Equipment capacity information

Levels of the Exercise

LEVEL 1: Basic Constraint Satisfaction (10 pts)

- Import data from Excel and set up the model in DOcplex.
- Create interval variables for each block-stage.

- Add feasibility constraints: one block per machine at a time, one stage per block.
- Deliverable: Feasible schedule (no optimization).

LEVEL 2: Precedence Constraints and Makespan Optimization (15 pts)

- Add precedence constraints (Stage 1 → Stage 2 → Stage 3 → ...).
- Define and minimize makespan.
- Deliverable: Optimized schedule with precedence and makespan.

LEVEL 3: Shared Resource Constraints (15 pts)

- Model cranes as a limited shared resource.
- Ensure no more operations use cranes simultaneously than available capacity.
- Deliverable: Model with resource constraints and impact analysis.

LEVEL 4: Setup Times (20 pts)

- Add sequence-dependent setup times between block types.
- Compare solutions with and without setups.
- Deliverable: Extended model with setup times and comparative results.

LEVEL 5: Alternative Objectives (15 pts)

- Introduce due dates for blocks.
- Minimize total tardiness and compare with makespan objective.
- Deliverable: Multi-objective analysis and comparison.

Deliverables

- **Code:** One Python script per level, clean and documented.
- **Report (8–10 pages):**
 - Problem approach and implementation by levels.
 - Key modeling decisions and results.
 - Comparison of solutions under different constraints/objectives.
 - Case insights and conclusions.
- **Presentation (10 min):**
 - Problem overview.
 - Main modeling steps.

- Key findings and recommendations.

Grading Rubric

Component	Points	Focus
Level 1: Feasibility	10	Basic model structure, constraint satisfaction
Level 2: Precedence + Makespan	15	Flow constraints, optimization objective
Level 3: Resource Constraints	15	Shared resource modeling, bottleneck analysis
Level 4: Setup Times	20	Sequence optimization, comparative analysis
Level 5: Alternative Objectives	15	Multi-objective understanding, trade-offs
Code Quality	10	Documentation, clarity, functionality
Report Quality	10	Analysis depth, insights, communication
Presentation	15	Clear communication, Q&A responses